

Systematic Review

Tax Fraud Detection Using Artificial Intelligence-Based Technologies: Trends and Implications

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Abstract

This study examines the role of artificial intelligence (AI) tools in enhancing tax fraud detection within the ambit of the OECD Tax Administration 3.0, focusing on how these technologies streamline the detection process through a new “Adaptive AI Tax Oversight” (AATO) framework. Through a textometric systematic review covering the period from 2014 to 2024, the integration of AI in tax fraud detection is explored. The methodology emphasizes the evaluation of AI’s predictive, analytical, and procedural benefits in identifying and combating tax fraud. The research underscores AI’s significant impact on increasing detection accuracy, predictive capabilities, and operational efficiency in tax administrations. Key findings reveal the ways by which the development and application of the AATO framework improves the tax fraud detection process. The implications highlight not only the governance benefits and ethical challenges that arise, but also provide practical guidance for tax authorities worldwide in leveraging AI to reduce compliance costs and strengthen regulatory frameworks. Finally, the study offers recommendations for future research, particularly in refining AI methodologies, differentiating policy implications across high-income and low- and middle-income countries, and addressing governance and ethical issues to ensure equitable and sustainable tax administration practices.

Keywords: artificial intelligence; tax fraud; AATO framework; blockchain; neural networks; data mining



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1. Introduction

Tax fraud is a global challenge with profound implications for societal trust and economic stability (Alm, 2023; Belahouaoui, 2025; De Roux et al., 2018). Defined as the intentional falsification of tax return information to illegally gain financial benefits or reduce tax liabilities, this phenomenon affects both high-income and low- and middle-income economies (Tax Justice Network, 2024). As a deliberate act of deception aimed at evading rightful tax payments, tax fraud not only deprives governments of essential revenue, but also undermines the integrity of tax systems (Gale & Krupkin, 2019; Murorunkwere et al., 2023a; Ruzgas et al., 2023). Given its widespread impact, recent international reports have sought to quantify the scale of tax fraud and its consequences for global public finances.

According to the Global Tax Evasion Report 2024, tax fraud continues to pose significant challenges to global tax revenues, despite international efforts to curb these practices. The report estimates that corporate tax revenue losses exceed USD 1 trillion annually, representing approximately 10 percent of global corporate tax revenues. These losses disproportionately affect non-Organization for Economic Co-operation and Development

(OECD) countries, where tax fraud strategies significantly erode public finances. Additionally, findings from the OECD Corporate Tax Statistics 2024 reveal that while corporate tax revenues represent 16 percent of total tax revenues globally, the share of corporate taxes as a percentage of Gross Domestic Product (GDP) varies widely across jurisdictions. For instance, in 2021, corporate tax revenues accounted for more than 25 percent of total tax revenues in 17 countries. These figures highlight the persistent disparities in the effects of profit shifting and the urgency of implementing more effective global tax policies to reduce these losses (Alstadsaeter et al., 2023; OECD, 2024; Belahouaoui & Attak, 2025).

Building on these global estimates, the Tax Justice Network (2024) provided more detailed insights into the distributional effects of tax abuse across income levels, estimating that global tax abuse results in an annual loss of USD 492 billion in tax revenues. Of this, USD 347.6 billion is lost due to corporate tax abuse by multinational companies, while USD 144.8 billion is attributed to offshore tax fraud by wealthy individuals. The financial burden is disproportionately heavier on lower-income countries, which lose USD 46 billion annually, equivalent to 36 percent of their public health budgets. In contrast, higher-income countries face an estimated loss of USD 446 billion per year, amounting to 7 percent of their public health budgets. If no significant policy changes are implemented, the Tax Justice Network (2024) projected that countries will collectively lose USD 4.92 trillion in tax revenue to tax havens over the next decade, exacerbating economic inequalities and weakening public services worldwide.

In response to these persistent challenges, the OECD has advanced its Tax Administration 3.0 vision, which seeks to harness digital transformation to strengthen tax systems. The OECD Tax Administration 3.0 vision aims for a thorough digital revamp of tax systems, both to respond to societal changes and also to harness digital advancements for reducing administrative burdens, fostering policy innovation, and curtailing tax fraud (OECD, 2020). This vision encapsulates the following six fundamental components: digital identity, taxpayer touchpoints, data management and standards, tax management and enforcement, the cultivation of new skill sets, and the establishment of governance frameworks. These components are pivotal for an effective, modern tax administration system. Reflecting on the Tax Technology Initiatives Inventory data, it is evident that this shift towards digitalization is substantial (OECD, 2023a, 2023b), underscoring the critical role of digital technologies in enhancing the operational efficiency and effectiveness of tax administrations, particularly in combating tax fraud (OECD, 2023a, 2023b).

In the context of Tax Administration 3.0, the integration of artificial intelligence (AI) tools plays a pivotal role in enhancing tax administration and benefiting taxpayers. AI tools, through advanced algorithms, improve liability assessment, enhance decision making, and provide an unprecedented efficiency in processing vast amounts of fiscal data, thereby reducing the need for human intervention and strengthening governments' oversight capacities. Recent studies illustrate how AI integration has advanced fraud detection. Gaie (2023) proposed a comprehensive approach to enhance e-Government and optimize fraud detection, while Mpofu (2024) highlighted the effectiveness of AI algorithms in identifying fraudulent activities. Tax et al. (2021) emphasized the critical role of machine learning within anti-fraud frameworks. Saragih et al. (2023) and Shakil and Tasnia (2022) showed how AI audits reduced fraud risks and improved tax collection efficiency. Relatedly, Reslan and Al Maalouf (2024) found that AI adoption in accounting had positive effects on financial reporting and fraud detection in accounting practices, and Louati et al. (2024) showed that AI adoption reduced fraud and enhanced security in blockchain transactions. More broadly, Bao et al. (2022) analyzed the wider implications of AI and big data for fraud detection, and Murorunkwere et al. (2022) demonstrated the effectiveness of neural networks in detecting income tax fraud in Rwanda. Collectively, these contributions

underscore the transformative impact of AI in building more secure and efficient tax systems (Belahouaoui, 2025).

Several recent studies have highlighted the need for further research to understand how artificial intelligence aids in detecting tax fraud, pointing out future directions for exploration in this field (Matheus et al., 2021; Tax et al., 2021; Zuiderwijk et al., 2021; Bassey et al., 2022; Bharosa, 2022; Gaie, 2023). These studies underscore the necessity for a comprehensive review over time, focusing on the benefits of AI and its role in enabling tax authorities to detect fraud in the era of digital transformation. Even so, this literature also suggests theoretical, methodological, and practical gaps.

This study addresses a significant research gap by clarifying the theoretical, methodological, and practical dimensions of AI-based technologies in tax fraud detection. Theoretically, although prior studies have acknowledged the potential of AI, there is still a lack of comprehensive understanding of which AI-based technologies are actually being deployed and how they contribute to the detection process. Methodologically, few systematic literature reviews have applied advanced analytical techniques—such as content analysis, lexicographic analysis, and textometry—supported by specialized textual analysis software, leaving an important gap in evidence synthesis. From a practical standpoint, existing studies do not sufficiently explore how the integration of AI into tax fraud detection can be made more effective through structured frameworks like the proposed “Adaptive AI Tax Oversight” (AATO). This study seeks to bridge these gaps by providing both a critical review of the literature and a conceptual framework to guide future applications.

Specifically, this study sets out to accomplish two main goals. First, we survey the AI-based tools predominantly used by global tax administrations for detecting tax fraud, focusing on their benefits and challenges. Second, we develop a conceptual framework detailing the application of artificial intelligence in tax fraud detection within the innovative landscape of the OECD Tax Administration 3.0. We attempt to not only map out the current utilization of AI in combating tax fraud, but also to guide the future integration of AI technologies in enhancing tax regulatory frameworks and operational efficiencies.

To meet these research objectives, we conduct a systematic literature review using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method, analyzing studies published between 2014 and 2024 from the Scopus and Web of Science databases. This comprehensive review involves a detailed textometric analysis of 163 studies, utilizing both bibliometric and textometric analyses of titles, abstracts, and keywords. Such approaches significantly enrich the depth and breadth of analysis in systematic literature reviews (Bueno et al., 2021). Furthermore, tools like IRAMUTEQ 0.7 alpha 2 are used for advanced statistical analysis of textual data, enhancing the quality of insights derived from the literature (Figura et al., 2023). These methodologies are not widely applied in systematic literature reviews, but they highlight existing gaps and present opportunities for generating more nuanced and extensive research findings in future studies. By adopting this combined bibliometric and textometric approach, this study constitutes the first systematic review to present the AI-based tools currently employed in tax fraud detection, to analyze the benefits and challenges associated with their use, and to propose a novel implementation framework that outlines the process of integrating these technologies into tax administrations. This underlines both the originality and significance of the study's contribution.

The paper is structured to provide a comprehensive exploration of the research. We start with the methods and data section, in which we detail the systematic review process and analytical approaches. We then present the results and discussions sections, first providing an initial descriptive analysis summarizing the broad findings, followed by

detailed results for an in-depth exploration. We conclude with a summary of our core insights and their broader implications.

2. Materials and Methods

Systematic literature reviews (SLRs) are celebrated for their structured and rigorous approach in consolidating existing research, crucially offering a comprehensive and unbiased perspective on studies concerning a specific subject (Boell & Cecez-Kecmanovic, 2015; Okoli, 2015). This method stands out in disciplines such as economics and business, aiding in the assimilation of varied methodologies amidst swiftly evolving and segmenting bodies of knowledge. Particularly for those delving into emerging fields like digital transformation and artificial intelligence (Aguilar et al., 2021; Guandalini, 2022), SLRs have proven to be an invaluable resource, both in spotting research voids and in establishing a robust groundwork for subsequent inquiries (Snyder, 2019).

We adopted the PRISMA method for its clarity and systematic approach in literature reviews. PRISMA provides a four-phase selection process, aiming to reduce bias and ensure comprehensive reporting (Liberati et al., 2009). We chose this methodology for its suitability to our research goals, allowing for a detailed and impartial exploration of the available literature in this evolving field (Page et al., 2021). Following PRISMA's guidelines facilitated a disciplined review process, culminating in a visual representation of the approach from study selection to inclusion. This method ensured the reliability and depth of the review, contributing to the understanding of how AI technologies help detect tax fraud in a variety of contexts.

2.1. Data Selection

We selected data from the following three primary databases: Scopus, Web of Science, and ScienceDirect, chosen for their high volume of quality publications, comprehensive coverage, advanced search capabilities, and reproducibility, making them the leading databases for systematic searches in this field (Mengist et al., 2020; Paul et al., 2021; Shaffril et al., 2021). This strategic combination of databases was aimed at achieving the most balanced search possible (Mourão et al., 2020), particularly valued for research in economics, management, and areas focusing on technological development and artificial intelligence. Our choice of these databases is supported by their efficiency and effectiveness in capturing the relevant literature in these specific domains (de la Torre-López et al., 2023; Gomes et al., 2022; Pranckutė, 2021). This methodological approach ensures a broad yet targeted review of the existing body of work, facilitating a comprehensive understanding of the subject matter.

The data collection process for this study followed the PRISMA research protocol (Figure 1) and adopted a multi-stage approach. In the identification phase, we conducted systematic searches across three major databases—Scopus, Web of Science, and ScienceDirect—using a comprehensive set of keywords combining terms related to “tax fraud detection”, “artificial intelligence”, “machine learning”, “neural networks”, and “big data”. This search initially yielded 215 records. In the screening phase, records were filtered according to publication period (2014–2024) and subject relevance, reducing the corpus to 195 articles. During the eligibility phase, non-English publications and inaccessible papers were excluded, resulting in 180 records. The remaining studies were assessed by carefully reviewing titles, abstracts, keywords, and full texts, which led to the identification of 168 potentially relevant articles. After applying our final inclusion and exclusion criteria—focusing on peer-reviewed studies directly addressing AI-based technologies in tax fraud detection and excluding papers that dealt solely with broader issues such as general digitalization or unrelated financial technologies—we retained 163 studies for the systematic literature

review. In addition, grey literature, including selected OECD reports and policy papers from international organizations, was consulted to provide further contextual insights and ensure comprehensive coverage.

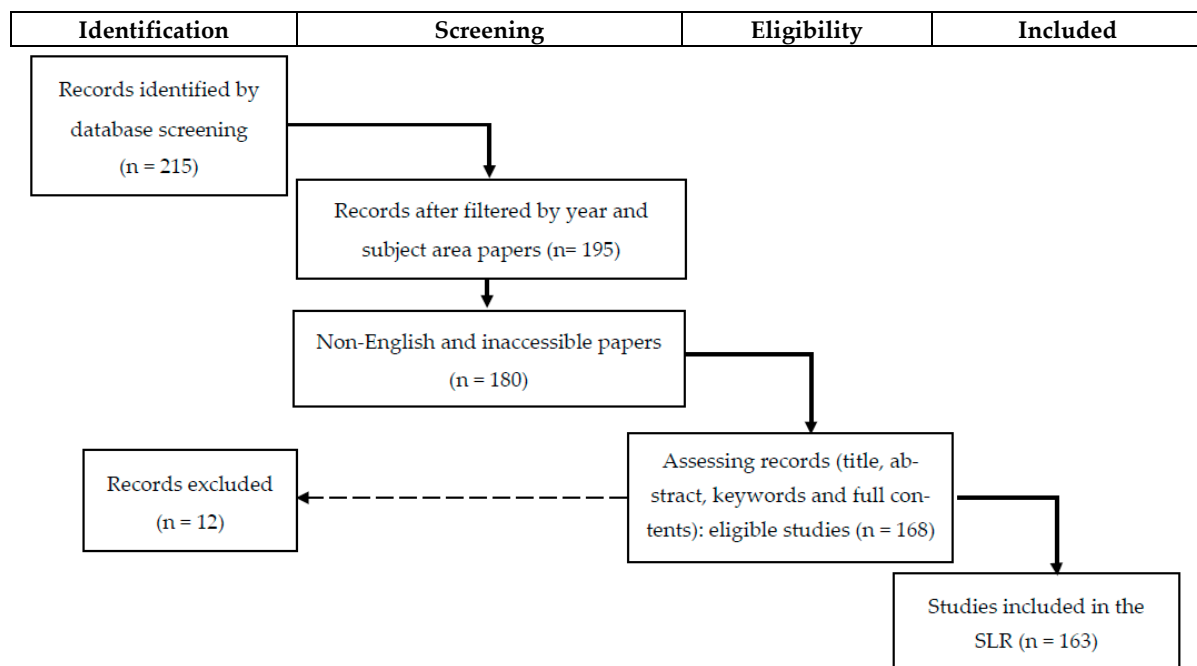


Figure 1. PRISMA research protocol.

The advanced search applied a precise combination of keywords and codes, as detailed in Table 1. Search strings combined “Artificial Intelligence” with terms related to tax administration, fraud, avoidance, evasion, and compliance. Keywords such as “Machine learning”, “Blockchain”, and “Neural networks” were part of the filter applied to the scanned items, which included article titles, abstracts, and keywords within the last decade (2014–2024).

Table 1. Keywords and codes used in advanced search.

Search strings	(“Artificial intelligence”) AND (“Tax administration” OR “Tax authorities”) AND (“Tax fraud” OR “Tax avoidance” OR “Tax evasion” OR “Tax compliance”)
Keywords filter	“Artificial intelligence” “Machine learning” “Blockchain” “Big data” “Neural networks” “Digital taxation” “Tax fraud” “Tax compliance” “Tax evasion” “Tax avoidance” “Tax administration” “Fraud detection” “Tax Systems”
Inclusion and exclusion criteria	The inclusion criteria retained peer-reviewed articles published between 2014 and 2024, written in English and explicitly addressing AI-based technologies in tax fraud detection, while the exclusion criteria eliminated non-English or inaccessible papers, studies outside the selected timeframe, and works unrelated to AI applications in fraud detection or lacking academic rigor
Scanned items	Article title, Abstract, Keywords
Year range	The last decade: 2014–2024
Database	Scopus, Web of Sciences and Science Direct

Source: Created by authors.

2.2. Data Analysis

We begin our data analysis with a bibliometric review to map the landscape of the research, covering publication trends over time (Figure 2), geographical contributions (Figure 3), the variety of paper types (Figure 4), and the range of subject areas within the dataset (Table 2). This step is critical for identifying patterns and establishing a contextual understanding of the field's development and focus areas regarding the use of AI for tax fraud detection.

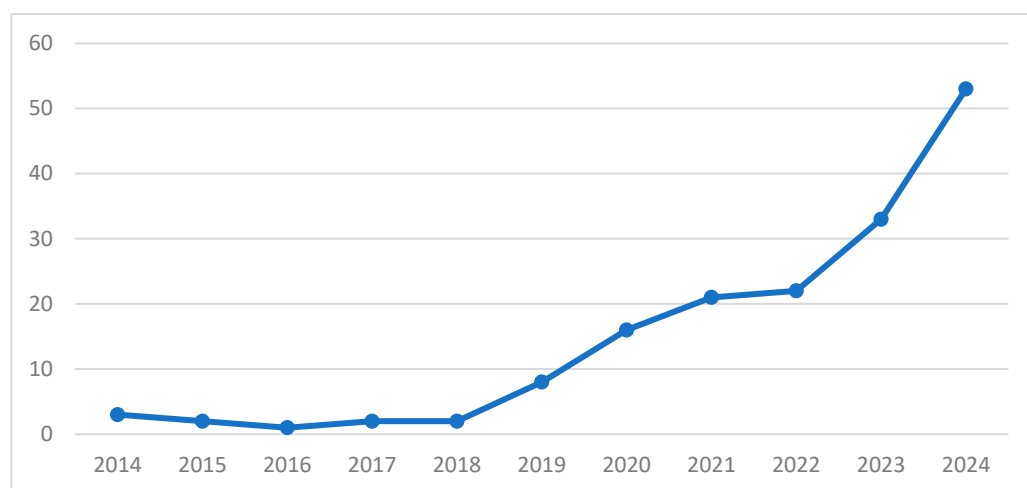


Figure 2. Documents by year. Source: Created by authors.

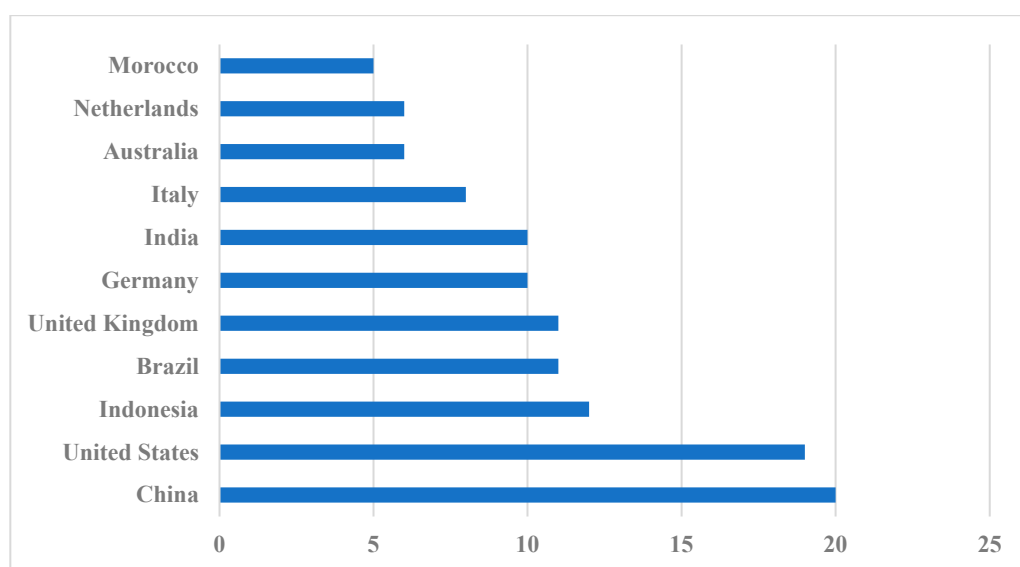


Figure 3. Country of affiliation. Source: Created by the authors.

Table 2. Documents by subject area.

Subject Area	Documents
Computer Science	98
Engineering	42
Business, Management and Accounting	37
Social Sciences	35
Economics, Econometrics and Finance	29
Decision Sciences	23
Materials Science	9

Source: Created by the authors.

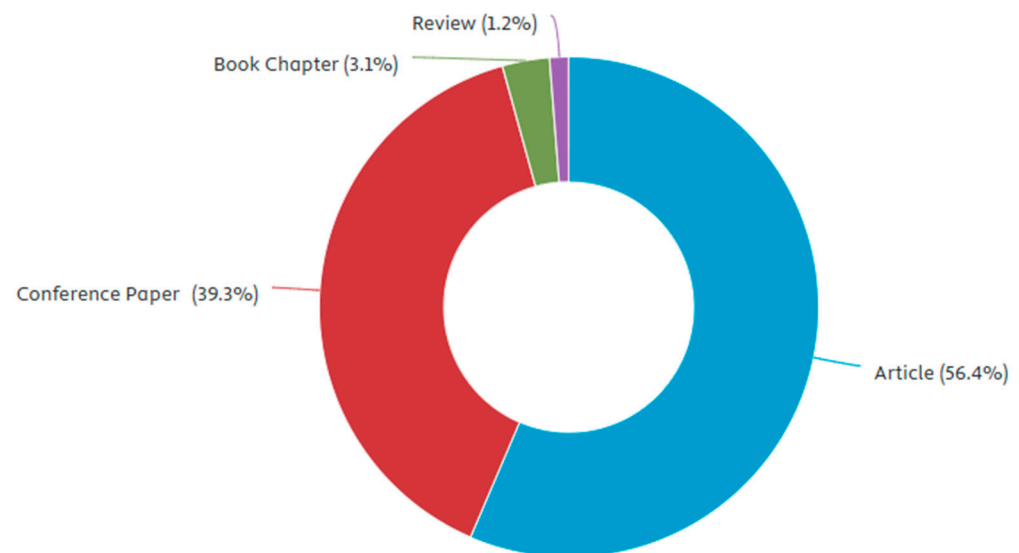


Figure 4. Type of papers. Source: Created by the authors.

Taking a more detailed approach to data analysis, we then conduct textometric analysis using IRAMUTEQ 0.7 alpha 2 software (Ratinaud, 2009), which allows for sophisticated analysis in line with the overarching goals of the study (Figura et al., 2023; Ramos et al., 2018). The choice of IRAMUTEQ 0.7 alpha 2 is justified by its ability to combine lexicographic, statistical, and textometric techniques within a single open-source platform, enabling rigorous, rapid, and replicable analysis of large corpora. Compared to other software such as NVivo and Leximancer, IRAMUTEQ 0.7 alpha 2 offers specific advantages for systematic reviews by automating similarity analyses and classification, thereby ensuring both depth and transparency in data interpretation (Figura et al., 2023). To reinforce the robustness of our findings, we also rely on triangulation and expert cross-checking of classifications to enhance reliability.

From the 163 papers gathered for the SLR, we create a general text corpus folder, containing the titles, abstracts, and keywords of the selected papers. We start with a global assessment, including a statistical summary (Table 3), identification of the 50 most frequently occurring words (Table 4), and a cluster dendrogram (Figure 5). We then conduct a detailed analysis that examines the 30 most important words per class (Table 5), performing factorial correspondence analyses (Figure 6) and conducting similarity analyses (Figure 7). These steps allow us to dissect the textual data thoroughly, providing a nuanced understanding of the prevalent themes and terms within the literature on AI and tax fraud detection.

Table 3. Statistical summary of corpus.

Concept	Total Number	
Number of occurrences	24 951	
Mean of occurrences per text	24 951	
Number of lexical forms (words)	3836	
Active forms	3607	
Number of clusters	3	6.44% of occurrences
Number of hapaxes legomenon (*)	1607	43.16% of forms

(*) Words with frequency = 1. Source: Created by the authors using IRAMUTEQ software.

Table 4. The 50 most frequently used active words.

Order	Active Forms	Freq.	Order	Active Forms	Freq.
01	Tax	285	26	Analytics	75
02	Machine learning	263	27	Challenges	75
03	Tax evasion	258	28	Companies	67
04	Taxation	219	29	Performance	63
05	Taxpayers	167	30	Identify	63
06	Model	158	31	Decision making	63
07	Fraud	152	32	Improve	62
08	Blockchain	150	33	Compliance	62
09	Artificial intelligence	147	34	Economic	58
10	System	125	35	Support	58
11	Approach	118	36	Context	58
12	Information	107	37	Management	55
13	Audit	105	38	Collection	54
14	Tax authorities	105	39	Neural networks	54
15	Tax fraud	100	40	Prediction	54
16	Tax administration	98	41	Digital	53
17	Methods	97	42	Accounting	52
18	Detection	95	43	Behavior	52
19	Fraud detection	95	44	Blockchain technology	51
20	Technology	88	45	Algorithm	51
21	Application	87	46	Big data	49
22	Data mining	87	47	Tax fraud detection	49
23	Process	81	48	Tax returns	48
24	Results	77	49	Value added tax	48
25	Government	77	50	Transformation	48

Source: Created by the authors using IRAMUTEQ software.

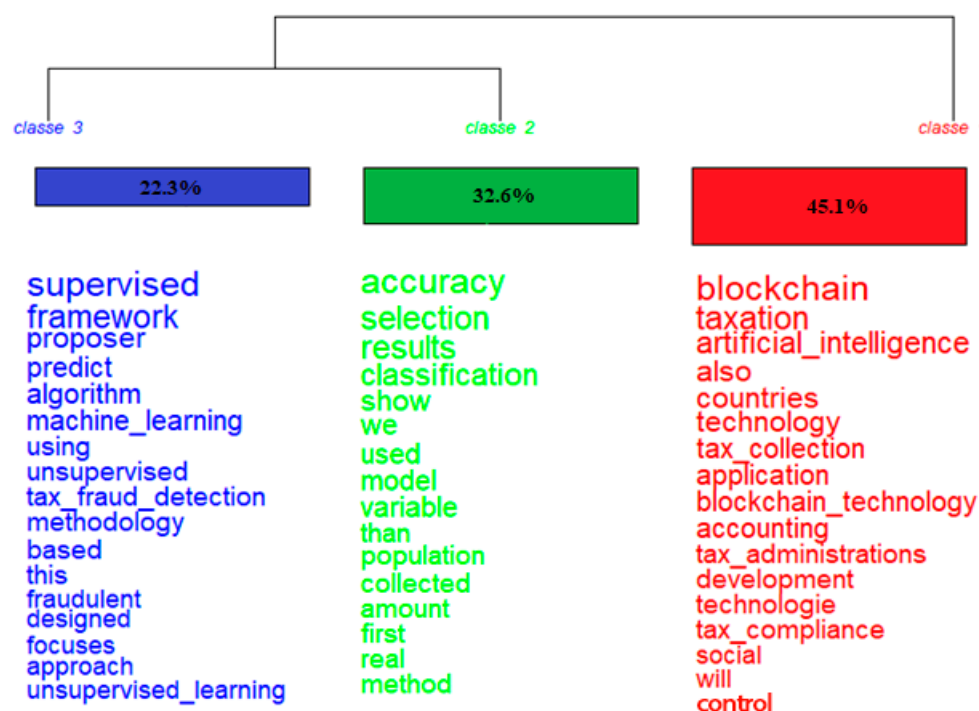
**Figure 5.** Dendrogram of the clusters. Source: Created by the authors using IRAMUTEQ software.

Table 5. The 25 most used words per class.

Cluster 1 (45%)—Blockchain and AI in Tax Modernization				Cluster 2 (33%)—Precision and Predictive Analysis in Tax Control				Cluster 3 (22%)—Learning Algorithms in Tax Fraud Detection			
Forms	Freq.	%	Chi2	Forms	Freq.	%	Chi2	Forms	Freq.	%	Chi2
Blockchain	48	83.33	31.14	Accuracy	19	94.74	34.52	Supervised	19	94.74	59.77
Taxation	32	87.50	24.72	Selection	21	85.71	27.98	Algorithm	24	79.17	46.99
Artificial intelligence	52	75.00	20.80	Structured	34	73.53	27.57	Machine learning	23	73.91	37.03
Tax collection	41	78.05	19.48	Networks	19	84.21	23.81	Tax fraud detection	11	90.91	30.57
Application	22	90.91	19.45	Detection	28	71.43	20.19	Fraudulent	13	84.62	29.92
Blockchain technology	35	80.00	18.43	Predictive models	113	49.56	18.58	Focuses	68	47.06	27.64
Accounting	13	100.0	16.23	Precision	48	60.42	18.47	Approach	82	43.90	26.17
Tax administrations	35	77.14	15.54	Estimate	69	55.07	18.09	Unsupervised learning	7	100.0	24.76
Development	15	93.33	14.51	Companies	11	90.91	17.33	Tax fraud	15	73.33	23.25
Technologies	15	93.33	14.51	Contributors	13	84.62	16.36	Boosting	11	81.82	23.01
Tax compliance	18	88.89	14.43	Investigate	10	90.00	15.24	Intelligent	70	44.29	22.54
Social	19	84.21	12.18	Audit	10	90.00	15.24	optimize	184	33.70	21.10
Control	13	92.31	12.00	Data mining	10	90.00	15.24	Fraud detection	8	87.50	19.97
Governance	29	75.86	11.73	False invoices	10	90.00	15.24	Predicting	8	87.50	19.97
Challenge	18	83.33	11.00	Taxable	19	73.68	15.08	Algorithms	8	87.50	19.97
Business	33	72.73	10.85	Efficiency	36	61.11	14.21	Crime	52	46.15	18.98
Processes	15	86.67	10.78	Dataset	9	88.89	13.16	Tax management	5	100.0	17.62
Analytics	12	91.67	10.76	Help	9	88.89	13.16	Supervised learning	5	100.0	17.62
Smart	12	91.67	10.76	Inspection	6	100.0	12.51	Deep learning	5	100.0	17.62
Legal	12	91.67	10.76	Estimation	6	100.0	12.51	Planning	13	69.23	16.98
Opportunities	28	75.00	10.68	Private	6	100.0	12.51	Neural network	11	72.73	16.52
Privacy	23	78.26	10.68	Unlabeled learning	33	60.61	12.49	Tax audit	7	85.71	16.50
Stakeholders	23	78.26	10.68	Tax avoidance	11	81.82	12.34	Tax returns	35	48.57	14.97
Principles	30	73.33	10.24	Resources	13	76.92	11.87	Big data	4	100.0	14.07
Mechanisms	20	80.00	10.23	Tax evasion	119	45.38	11.25	Personal income tax	4	100.0	14.07
International	8	100.0	9.89	Neural networks	8	87.50	11.11	ML Algorithms	4	100.0	14.07
Modernization	48	83.33	31.14	Declarations	8	87.50	11.11	Decision making	4	100.0	14.07
Perceived	32	87.50	24.72	Tax assessment	5	100.0	10.41	Data analytics	4	100.0	14.07
Protection	52	75.00	20.80	Active learning	5	100.0	10.41	Corporate tax	4	100.0	14.07
Costs	41	78.05	19.48	Vertical equity	10	80.00	10.39	Data mining	4	100.0	14.07

Source: Created by the authors using IRAMUTEQ software.

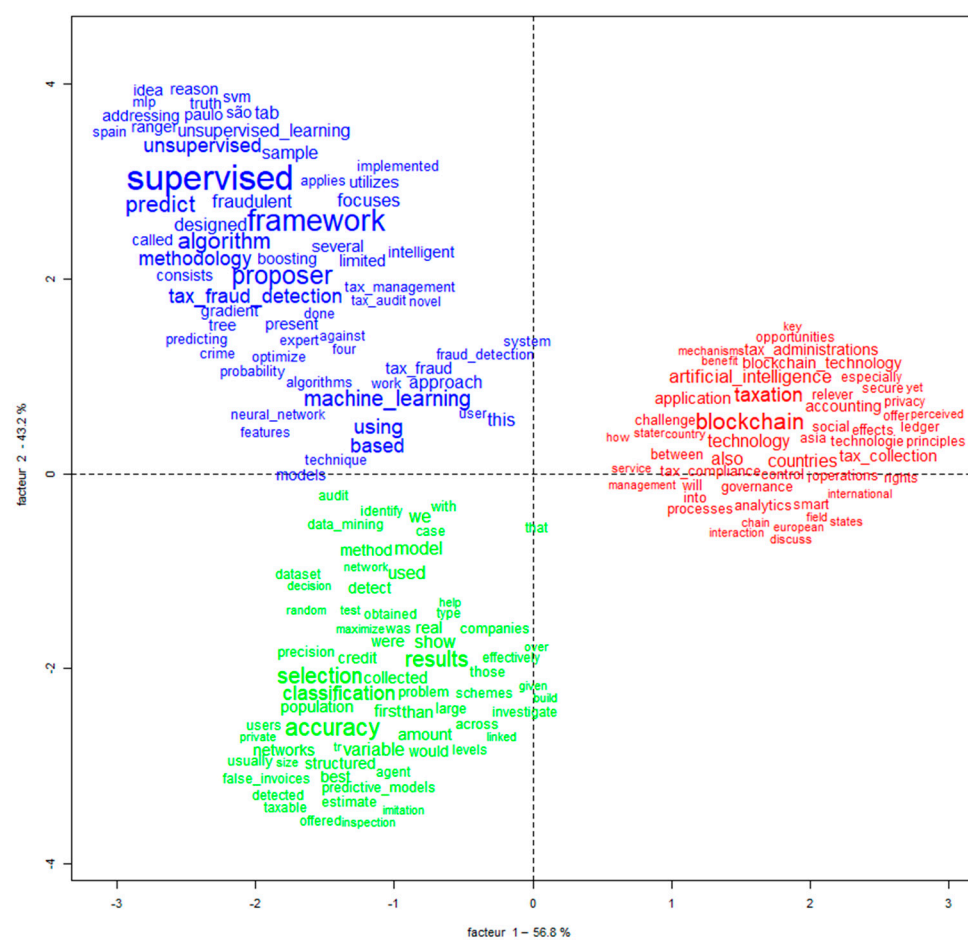


Figure 6. Factorial correspondence analyses. Source: Created by the authors using IRAMUTEQ software.

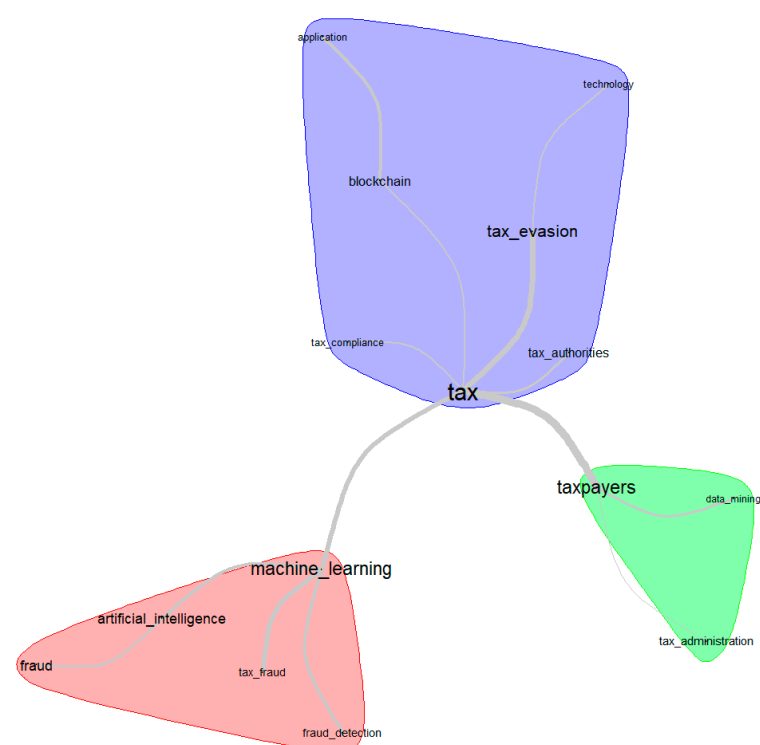


Figure 7. Similarity analysis. Source: Created by the authors using IRAMUTEQ software.

3. Results (1): Descriptive and Bibliometric Analysis

In this section, we conduct descriptive and bibliometric analyses to provide a foundational overview of AI research in tax fraud detection, including trends, contributions, and geographic insights. Figure 2 illustrates the number of publications by year, showing a continuous increase in research output over time. Notably, there has been a sharp rise in recent years, with 53 documents in 2024, 33 in 2023, 22 in 2022, and 21 in 2021. This upward trend underscores a growing scholarly focus on AI in tax fraud detection, likely driven by its increasing relevance and the need for advanced methodologies to address evolving economic and regulatory challenges.

Figure 3 indicates that research contributions on AI in tax fraud detection are geographically diverse, with China ($n = 20$) leading in the number of documents published, closely followed by the United States ($n = 19$). Indonesia ($n = 12$) then follows, reflecting a significant body of work emanating from this country as well. Brazil and the United Kingdom are tied ($n = 11$ each), also contributing to the research output. This spread suggests a wide international interest and investment in exploring AI applications for tax fraud detection, indicating the global relevance of the issue.

These many papers are predominantly (Figure 5) made up of articles (51.9 percent) and conference papers (44.4 percent), indicating a strong representation in both formal publications and academic gatherings. Book chapters and reviews each constitute a small fraction (1.9 percent), suggesting a lesser focus on these formats within the current discourse on AI in tax fraud detection. This distribution underscores the research community's preference for disseminating findings through articles and presentations at conferences.

The subjects addressed in the collected documents predominantly fall under Computer Science, indicating its central role in the discussion of AI for tax fraud detection. Other fields such as Engineering, Business, Management and Accounting, and Social Sciences also contribute significantly, demonstrating the interdisciplinary nature of this research (see Table 2). Contributions from Economics, Econometrics and Finance, Decision Sciences, and even Materials Science reflect the broad relevance and application of AI across various facets of research connected to tax fraud.

The bibliometric analysis reveals an upsurge in AI and tax fraud detection research, especially in recent years, with a global spread of contributions led by China and the United States. Computer Science dominates among the subject areas, with interdisciplinary input from various fields, illustrating the wide-ranging impact of AI in enhancing tax fraud detection and the diverse academic interest it garners.

We now engage in text analysis, extracting key findings in order to capture essential insights from the exploration of AI in tax fraud detection.

The statistical summary provided in Table 3, generated using IRAMUTEQ software, offers insightful metrics on the corpus post lemmatization. The table highlights a total of 24,951 occurrences across the corpus, with an average of 3836 occurrences per text and a total of 3607 lexical forms (words) identified. Among these, only 3 forms are categorized as active, with 1607 being unique occurrences (hapaxes legomenon), signifying a diverse range of vocabulary with a substantial portion being used only once. This diversity and the ratio of unique terms to total occurrences (6.44 percent) and forms (43.16 percent) suggest a wide breadth of topics and concepts covered within the texts analyzed. Such detailed analysis underscores the complexity and richness of the language surrounding AI in tax fraud detection research.

Table 4 showcases the 50 most frequently used active words in the corpus, illustrating key focus areas within AI and tax fraud detection research. Terms like "tax", "machine learning", "tax evasion", and "blockchain" are among the most used words, reflecting the core subjects and technologies under investigation (Alm, 2021; Alm et al., 2019). The

prominence of “artificial intelligence” alongside specific methods like “data mining” and “neural networks” underscores the technological emphasis of the studies (De Roux et al., 2018; Delgado et al., 2023; Murorunkwere et al., 2022; Pérez López et al., 2019). Additionally, the inclusion of “fraud detection”, “tax authorities”, and “government” highlights the practical applications and stakeholders involved. This frequency analysis not only identifies the primary topics and tools explored, but also indicates an interdisciplinary approach encompassing technology, regulatory concerns, and the broader economic and management implications in addressing tax fraud.

The dendrogram in Figure 6, created by IRAMUTEQ software, illustrates the clustering of concepts within the corpus. The three distinct classes show a thematic division. Class 1, the largest group, focuses on foundational technologies and their application in the field, with keywords like blockchain, taxation, and artificial intelligence (Mazur, 2022; Morton & Curran, 2023). Class 2 centers on research methodologies, with terms like accuracy and classification indicating a focus on the precision and categorization aspects of the studies (Murorunkwere et al., 2022). Class 3 addresses specific types of machine learning, highlighted by terms such as supervised and unsupervised learning, suggesting an emphasis on the algorithms used in tax fraud detection (Alm et al., 2019; De Roux et al., 2018; Savić et al., 2022). Each cluster represents a significant area of focus in the literature, providing a visual breakdown of the key topics and approaches in the use of AI for detecting tax fraud (Table 5).

The first cluster in Table 5 centers on “Blockchain and AI in tax modernization”, and it is characterized by a strong focus on the integration of blockchain technology and artificial intelligence within the domain of taxation (Fatz et al., 2019; Mazur, 2022; Morton & Curran, 2023; Owens & Hodžić, 2022). This theme reflects an intersection between advanced technological applications and their role in transforming tax collection, administration, and compliance strategies (Baghdasaryan et al., 2022; Cobham & Janský, 2018; Martínez et al., 2022). The presence of terms such as blockchain, tax collection, accounting, and tax compliance suggests an exploration of how these digital tools can improve the efficiency, transparency, and effectiveness of tax systems (Baghdasaryan et al., 2022; Martínez et al., 2022). It also implies a consideration of the challenges, opportunities, and legal aspects related to the adoption of these technologies in an international context, indicating a comprehensive discussion on the modernization of tax regimes through tech advancements. For Cluster 2, which makes up 33 percent of the focus, a suitable theme might be “Precision and predictive analysis in tax control”. This cluster is dominated by terms like accuracy, selection, and structured, which align with the precision required for effective tax monitoring and the use of predictive models for foresight in tax-related issues (Sampa & Phiri, 2023; Xu et al., 2023). Additionally, the cluster reflects on the importance of detection, data mining, and audit, all of which are crucial for identifying and addressing tax evasion and avoidance with accuracy (De Roux et al., 2018; Sampa & Phiri, 2023). These terms, along with efficiency and resources, suggest a focus on optimizing the tax assessment process and improving the accuracy of identifying tax-related discrepancies. Cluster 3 accounts for 22 percent of the data, and could be themed as “Learning algorithms in tax fraud detection”. It is characterized by a focus on supervised and unsupervised learning methods, as well as the utilization of specific algorithms and machine learning techniques tailored for tax fraud detection (Masrom et al., 2022; Murorunkwere et al., 2023a, 2023b; Savić et al., 2022). The cluster also signifies an emphasis on fraudulent behaviors and the approaches developed to predict and identify them within tax systems. This cluster suggests an analysis of the effectiveness of different AI methodologies, such as deep learning and neural networks, in enhancing the precision of fraud detection and tax audits (Baghdasaryan et al., 2022).

The factorial correspondence analysis in Figure 6 visualizes the relationship and correlation between the various keywords used in the literature on AI and tax fraud detection. In this graph, the closer the words are to each other, the more frequently they are associated within the documents. It is clear from the dense clustering of terms like supervised, framework, and predict that there is a strong focus on predictive models and structured approaches in AI. Another dense cluster includes blockchain, taxation, and artificial intelligence, highlighting the technological convergence in this research area. This type of analysis allows us to observe how different concepts within the field are interconnected, offering a graphic representation of the thematic structures that underpin the research corpus.

Figure 7 presents a similarity analysis, which visually groups related concepts in the field of AI and tax fraud detection into clusters. These clusters represent interconnected themes, with one focusing on the core subject of tax, surrounded by related terms like evasion and compliance, another emphasizing machine learning and its link to fraud and fraud detection, and a third highlighting taxpayers and associated aspects such as data mining and tax administration (Belahouaoui & Attak, 2024b). Each cluster is shaped by the frequency and co-occurrence of terms in the literature, indicating how different topics are conceptually related to each other within the research field.

The text analysis crystallizes the focus on cutting-edge AI technologies and methods like blockchain and machine learning in enhancing tax fraud detection, as reflected in the literature's predominant themes and the clustering of related research topics.

4. Results (2): Key Findings

In this section, we examine the key findings from a carefully selected set of studies, highlighting major contributions and challenges in the use of AI for tax fraud detection. We chose the ten studies presented in Table 6 by using a comprehensive literature analysis, considering citation impact, relevance to the research objectives, and thematic alignment. Selection criteria focused on the most cited studies, ensuring strong methodological foundations and direct connections to AI-driven tax fraud detection. This synthesis provides essential perspectives on the effectiveness, limitations, and future developments of AI in tax administration.

Table 6. Key findings from the top 10 most cited studies directly related to the subject.

Authors	Purpose	Variables	Results
Calafato et al. (2016)	To improve the process of identifying tax fraud using a controlled natural language.	• Efficiency of fraud detection patterns; • Ease of use for fraud experts	Enables fraud experts to optimize rules swiftly and independently, reducing the time and errors associated with traditional methods.
Alm et al. (2019)	To analyze the impact of technological advancements on tax compliance, exploring both enhancements and challenges posed by emerging digital tools.	• Technological advancements • AI, data mining, blockchain • Public-key cryptography • Tax compliance and noncompliance	Technological advancements improve tax compliance by aiding enforcement (AI, data mining, cryptography).

Table 6. Cont.

Authors	Purpose	Variables	Results
Savić et al. (2022)	To enhance tax evasion risk management through a novel hybrid unsupervised method.	• Tax evasion • Outlier detection • Unsupervised learning • Clustering • Representational learning	Demonstrates a high efficacy in identifying internally validated outliers (90–98%), improving outlier interpretability and relevance in economic contexts.
Baghdasaryan et al. (2022)	To enhance tax audit efficiency in Armenia through a machine learning model, particularly for businesses under a standard tax regime.	• Historical fraud • Audit data • Administrative cost shares, • External economic activities	Demonstrates the potential to improve fraud detection accuracy over rule-based methods and highlights the value of taxpayer network data, especially for new companies lacking historical fraud data.
Murorunkwere et al. (2022)	To identify factors contributing to income tax fraud using Artificial Neural Networks (ANNs).	• Fraud detection • Income tax • Multilayer perceptron • Neural network • Tax fraud	Achieves a high performance in fraud detection with 92% accuracy, 85% precision, 99% recall, and 95% AUC-ROC. Identifies specific business-related factors as relevant to income tax fraud.
Alsadhan (2023)	To address limitations in current tax fraud detection methods by proposing a comprehensive framework combining supervised and unsupervised machine learning models.	• Tax fraud • Feature engineering • Applied machine learning	Demonstrates the framework's effectiveness in identifying potential tax fraud using data from the Saudi tax authority.
Prolhac and Gaie (2023)	To introduce a novel framework for optimizing tax fraud detection.	• Artificial intelligence • E-government • Fraud detection • Neural networks • Open framework	The framework enhances fraud detection efficiency and is made publicly available for further research and optimization.
Ruzgas et al. (2023)	To enhance the efficiency of tax evasion detection in Lithuania using data mining techniques.	• Tax evasion • Fraud detection • Data mining, clustering • Prediction	Demonstrates the effectiveness of data mining in detecting tax evasion, offering tools for reducing revenue losses and aiding decision makers in low- and middle-income countries.

Table 6. Cont.

Authors	Purpose	Variables	Results
Belahouaoui and Attak (2024a)	To analyze the impact of tax digitalization, focusing on AI technologies, on enhancing tax compliance behavior in the context of Tax Administration 3.0.	• Tax Administration 3.0 • Tax compliance • Digital taxation • Artificial intelligence	AI and blockchain significantly improve tax compliance and efficiency. Challenges persist in emerging economies regarding adoption and integration. The trend toward Tax Administration 3.0 highlights the need for regulatory frameworks and SME support.
Khaltar (2024)	To examine the impact of governance quality and adoption of the Open Government Partnership on trade-related tax evasion in low- and middle-income countries.	• Tax evasion • Low- and middle-income countries • Governance quality • Open government partnership	Governance quality and open government initiatives significantly reduce trade-related tax evasion. Open government adoption moderates the effect of governance quality, strengthening its impact on reducing tax evasion.

Source: Created by the authors.

To further elucidate the practical applications and implications of various AI tools in the realm of tax fraud detection, we present a detailed comparative analysis in Table 7. This table synthesizes essential aspects of each AI technology, including their specific applications in detecting tax fraud, the benefits they offer, the challenges they pose, and the associated costs. This structured overview serves as a comprehensive guide for tax authorities and policymakers to assess the suitability of different AI tools in enhancing their fraud detection capabilities, balancing technological advantages against operational challenges and cost considerations.

In this analysis, we explore how AI tools function in a tax administration context, focusing on their effectiveness, efficiency, cost implications, and the challenges they pose. Machine learning and deep learning stand out for their pattern recognition capabilities, essential for sifting through complex taxpayer data to spot subtle fraud indicators. The resource demands of each technology, including computational and human resources, are assessed, highlighting how technologies like data analytics and predictive modeling streamline the fraud detection process by minimizing the need for extensive manual labor. Moreover, the financial aspects of implementing AI solutions are discussed. Blockchain technology, for instance, provides transparency and security but comes with high setup and ongoing maintenance costs. Additionally, the analysis tackles the hurdles that each technology might face, such as data privacy concerns associated with big data applications and the steep learning curve associated with neural networks, which often require specialized knowledge and training. This comparative overview aims to arm tax authorities and policymakers with information to make informed decisions regarding the integration of AI tools into their systems, identifying which tools best align with their specific needs by balancing effectiveness, efficiency, and cost to optimize their tax fraud detection approach.

Table 7. AI tools used to detect tax fraud.

AI Tools	Tax Fraud Detection	Benefits	Challenges	Costs
Machine Learning	Identifying patterns and anomalies in large datasets.	High accuracy and adaptability to new fraud patterns.	Requires large, clean datasets for training.	High computational resources for training and running models.
Deep Learning	Deep analysis of complex and layered data structures.	Excellent at processing large volumes of unstructured data.	Complex to configure and requires extensive training data.	High initial setup and operational costs.
Big Data	Handling and processing vast datasets to find irregularities.	Can manage and analyze data at a scale beyond human capabilities.	Privacy concerns; requires robust data governance.	Significant infrastructure and storage costs.
Data Mining	Discovering patterns and correlations in large datasets.	Uncovers hidden patterns that might indicate fraud.	Can produce many false positives without proper tuning.	Requires investment in data processing tools and technology.
Data Analytics	Analyzing taxpayer data to predict and identify fraudulent transactions.	Enables real-time decision making and trend analysis.	Needs skilled personnel to interpret data correctly.	Costs associated with analytical tools and personnel training.
Blockchain	Providing a secure and transparent record of transactions.	Increases data integrity and security.	Technologically complex and requires consensus on data entry.	High implementation and maintenance costs.
Predictive Modeling	Forecasting future trends based on historical data.	Allows proactive measures against predicted fraud activities.	Models can be inaccurate if data or assumptions are not correct.	Development and continual update costs.
Neural Networks	Learning and recognizing complex patterns of tax fraud.	Extremely effective at identifying subtle patterns.	Requires large amounts of training data and computing power.	High costs for setup, operation, and maintenance.
Unlabeled Learning	Learning from data that has not been explicitly labeled as fraudulent or non-fraudulent.	Useful in scenarios with limited labeled data.	Less accurate than supervised learning models.	Computational costs for processing and model tuning.
Active Learning	Iteratively querying a user to label data points.	Improves model accuracy with fewer training data.	Depends on continuous user interaction for labeling.	Costs of setup and iterative process involvement.
Algorithm	General computational methods for detecting tax fraud.	Broad applicability and customizable to specific needs.	Algorithmic bias and transparency issues.	Development, testing, and deployment costs.

Source: Created by the authors.

The ten most cited studies highlight the strong potential of AI-based technologies, such as machine learning, neural networks, and data mining, for improving tax fraud detection, often achieving high accuracy rates and efficiency gains. However, these contributions remain fragmented and heavily context-specific, limiting the generalizability of the findings.

Most studies focus primarily on technical performance while overlooking institutional, regulatory, and ethical challenges, including costs, data privacy, and skill shortages. Moreover, the link between technological effectiveness and broader outcomes such as taxpayer trust and compliance remains underexplored. This indicates a clear need for more integrative and critical research that connects technological innovation with governance, equity, and sustainability in tax administration.

To provide a clearer understanding of how different AI technologies operate in tax fraud detection, Table 7 presents a structured comparison of the main tools identified in the literature.

While Table 7 provides a comparative overview of AI tools used in tax fraud detection, their adoption and effectiveness are not solely determined by technological performance. According to the literature, several categories of factors influence the implementation of AI-based technologies in tax administrations. Table 8 presents these categories and specifies the key factors within each, highlighting the interplay between technological, organizational, institutional, and socio-economic dimensions.

Table 8. Factors influencing AI adoption in tax fraud detection.

Category of Factors	Specific Factors
Technological	Data availability and quality; computational resources; integration with existing systems; scalability of solutions.
Organizational	Staff skills and expertise; training and capacity building; management support; change management readiness.
Institutional and Regulatory	Legal frameworks; data privacy regulations; cybersecurity standards; ethical and governance considerations.
Economic and Financial	Implementation costs; maintenance and infrastructure expenses; return on investment (ROI); budget constraints in public administrations.
Socio-political	Public trust in AI systems; transparency and accountability; international cooperation; alignment with digital government strategies.

Source: Created by the authors.

Building on the bibliometric and textometric analyses, we now present the “Adaptive AI Tax Oversight” (AATO) framework, depicted in Figure 8. Rather than being a purely descriptive taxonomy, the framework derives directly from the systematic synthesis of the most cited studies and clusters identified through our analyses, thereby linking empirical evidence with theoretical reflection. The AATO framework integrates key AI functions—data aggregation, anomaly detection, predictive analysis, adaptive learning, decision support, and continuous evolution—into a dynamic process that resonates with broader models of digital governance, institutional theory, and public choice approaches. By grounding its components in both the literature review and established theoretical perspectives, the framework not only validates the themes that emerged from the bibliometric findings, but also provides practical implications for tax authorities. Specifically, it offers a structured guide for leveraging AI within the OECD’s Tax Administration 3.0 vision, helping policymakers move beyond descriptive discussions toward actionable strategies that balance technological innovation, institutional capacity, and governance principles in combating tax fraud.

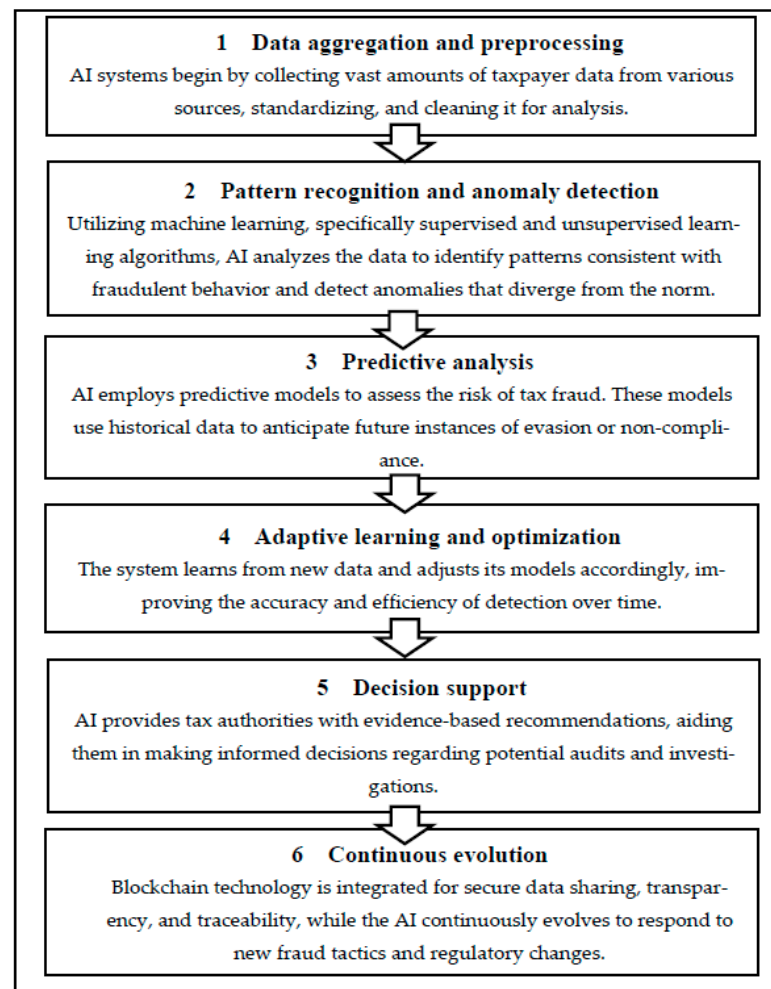


Figure 8. Adaptive AI Tax Oversight (AATO) framework. Source: Created by the authors.

Delving into the intricacies of Tax Administration 3.0, it is essential to examine both the advantages and challenges of deploying AI in tax fraud detection. Table 8 attempts to present a balanced view, juxtaposing the benefits that AI brings to the efficiency and effectiveness of tax fraud detection against the challenges it poses in terms of technical implementation, data management, and ethical considerations. This dichotomy is crucial for understanding the full spectrum of AI's impact on modern tax administration practices.

Table 9 aims to summarize the potential advantages and obstacles that come with integrating AI into tax fraud detection processes, highlighting considerations for tax administrations moving towards a more digital and AI-driven future.

A critical analysis of the reviewed studies indicates that while AI applications in tax fraud detection demonstrate considerable potential, most contributions remain short-term and highly context-specific, with limited integration into broader theoretical or institutional frameworks. The literature often emphasizes technical accuracy and efficiency, yet pays insufficient attention to long-term trends such as the scalability of AI tools, the sustainability of their integration within tax administrations, and their impact on taxpayer trust and compliance. Moreover, the key factors influencing adoption, ranging from data quality and technological infrastructure to governance, ethical safeguards, and organizational readiness, are frequently underexplored or treated in isolation. Addressing these dimensions more explicitly would not only strengthen the theoretical contributions of the field by linking AI adoption to models of digital governance and institutional theory, but also generate more practical insights for policymakers seeking the sustainable and equitable integration of AI into tax fraud detection systems.

Table 9. Benefits and challenges of AI in tax fraud detection within the framework of Tax Administration 3.0.

Benefits of AI in Tax Fraud Detection		Challenges of AI in Tax Fraud Detection	
Enhanced accuracy	AI algorithms can analyze vast datasets with precision, reducing human errors and increasing the accuracy of fraud detection.	Data privacy concerns	The use of AI requires the handling of large volumes of personal data, raising concerns about privacy and data protection.
Predictive capabilities	AI's predictive models can forecast potential fraud cases by learning from historical patterns.	Complexity of tax evasion tactics	As fraudsters evolve their tactics, AI systems must constantly adapt, which can be technically challenging.
Resource efficiency	Automating the detection process with AI allows tax authorities to allocate human resources more strategically.	Integration with existing systems	Incorporating AI into the legacy systems of tax administrations can be technically and financially demanding.
Real-time analysis	AI enables continuous monitoring and real-time analysis, leading to quicker responses to fraudulent activities.	Need for expertise	There is a demand for skilled professionals to develop, manage, and interpret AI systems, which can be a scarcity.
Scalability	AI systems can be scaled to handle the increasing volume and complexity of tax data as administrations grow.	Legal issues	The decision-making process of AI must align with legal standards and ethical considerations, which can be complex to navigate.
Transparency and traceability	Blockchain integrated with AI can provide a transparent and traceable record of transactions.	Algorithm bias	AI models can inadvertently perpetuate biases present in the training data, leading to unfair or skewed outcomes.
Governance	AI strengthens transparency, accountability, and trust in tax administration while supporting evidence-based policymaking.	Ethical issues	AI raises risks of algorithmic bias and discrimination, along with concerns about data privacy, surveillance, and the protection of taxpayer rights.

Source: Created by the authors.

5. Conclusions

This study contributes to the literature on AI and tax fraud detection by providing one of the first systematic reviews that integrates bibliometric and textometric approaches. Theoretically, it clarifies how AI-based technologies such as machine learning, neural networks, and blockchain are positioned within the evolving framework of Tax Administration 3.0, highlighting both their potential and their limitations. Methodologically, the adoption of a combined bibliometric–textometric strategy enriches evidence synthesis and offers a replicable model for future research in the field. Practically, the development of the Adaptive AI Tax Oversight (AATO) framework provides policymakers and tax authorities with a structured guide to assess the suitability of AI tools, balancing efficiency gains with governance, ethical, and financial considerations. By addressing theoretical, methodological,

and practical gaps, the study advances the academic debate while also offering actionable insights for the sustainable integration of AI into tax administration systems.

The insights derived from the reviewed studies were synthesized into the AATO framework (Figure 8). This framework outlines a structured process in which AI technologies support tax fraud detection, progressing from data aggregation to pattern recognition, predictive analysis, adaptive learning, decision support, and blockchain-enabled transparency. By integrating supervised and unsupervised learning with secure distributed ledgers, the model illustrates how AI can transform tax administration into a more dynamic, accurate, and efficient system. Nonetheless, the findings also underscore the dual nature of AI in this domain: while offering scalability, predictive capacity, and real-time monitoring, significant challenges persist. These include data privacy concerns, integration barriers with legacy systems, the need for highly skilled expertise, and ethical risks such as algorithmic bias.

At the same time, the study acknowledges that the technical achievements of introducing AI have been more extensively addressed than their broader consequences. The causal connection between technological performance and social or institutional outcomes such as taxpayer trust, institutional change, and governance remains insufficiently explained. This gap calls for future research to explicitly examine how the adoption of AI in tax administrations translates into long-term changes in institutional practices, public trust, and the legitimacy of tax systems. The AATO framework, therefore, emphasizes not only the potential of AI to modernize tax fraud detection, but also the necessity of robust governance mechanisms, ethical safeguards, and institutional adaptation to ensure its sustainable adoption.

5.1. Implications of the Study

5.1.1. Theoretical Implications

This study contributes to the theoretical advancement of research on AI-driven tax fraud detection by deepening the understanding of how different methodological approaches, particularly supervised and unsupervised learning, interact to address the complexity of tax evasion. Beyond enriching the existing literature on taxation and technology, it highlights the potential of AI to serve as a bridge between predictive modeling and adaptive oversight in fiscal systems. From a broader perspective, the study underscores the importance of adopting an interdisciplinary lens, combining insights from data science, legal scholarship, and economic theory. Such a convergence not only strengthens the conceptual foundations of AI applications in tax administration, but also provides a theoretical basis for developing more robust, scalable, and context-sensitive frameworks. By doing so, the research opens avenues for future inquiries into how AI can be theoretically integrated with established models of governance, institutional theory, and public choice, thereby embedding technological innovation within recognized academic paradigms.

5.1.2. Implications for Governments and Tax Administrations

The findings of this study carry significant implications for governments and tax administrations, with recommendations that vary according to the level of technological maturity of each context.

In advanced economies, where tax administrations are already supported by sophisticated digital infrastructures, the priority lies in enhancing interoperability, strengthening ethical governance, and addressing challenges related to algorithmic bias and data privacy. These administrations should also focus on integrating AI into predictive and real-time fraud detection systems, while aligning with broader digital governance frameworks to maintain transparency and taxpayer trust.

In contrast, for low- and middle-income countries, the emphasis should first be placed on building robust digital infrastructures and ensuring access to reliable and standardized data. Gradual adoption of AI tools is recommended, beginning with low-cost data analytics and supervised learning models before transitioning toward more complex technologies such as blockchain and deep learning. Policymakers in these contexts must also prioritize capacity-building programs, including specialized training for tax officials, partnerships with universities and technology providers, and phased regulatory reforms. Such incremental approaches can help mitigate implementation risks while allowing administrations to progressively harness the benefits of AI.

Taken together, these differentiated policy implications highlight the need for context-sensitive strategies: advanced economies must refine and regulate AI applications, while low- and middle-income countries must first establish foundational capacities to support their sustainable integration. This dual perspective ensures that AI-driven tax fraud detection becomes a feasible and equitable instrument across diverse institutional and economic environments.

5.1.3. Implications for Audit Companies and Professionals

For audit companies and professionals, the adoption of AI in tax fraud detection brings significant transformations. On the positive side, AI strengthens audit quality by enhancing the detection of irregularities, improving risk-based approaches, and allowing real-time monitoring of complex financial operations. These technologies also enable auditors to shift from routine manual tasks toward higher value-added analytical and advisory roles. However, the integration of AI also poses challenges: professionals must continuously update their digital, statistical, and legal knowledge, adapt to evolving technologies, and ensure the ethical use of algorithms. Maintaining professional judgment and independence becomes essential to avoid over-reliance on automated systems, highlighting the need for continuous training and ethical guidelines in the audit profession.

5.1.4. Implications for Taxpayers

For taxpayers, AI-driven tax fraud detection systems have the potential to reinforce fairness, transparency, and consistency in tax administration, thereby improving overall trust in public institutions. Automated processes may reduce discretionary errors and ensure that similar cases are treated equally. At the same time, taxpayers face new concerns related to data privacy, surveillance, and the risk of algorithmic mistakes that could unfairly classify compliant taxpayers as fraudulent. Ensuring clear communication, legal safeguards, and accessible channels for contesting automated decisions is, therefore, crucial. These measures will help balance efficiency gains with the protection of taxpayers' rights, fostering greater acceptance of AI-based tax systems.

5.2. Limitations of the Study

This study presents certain limitations that should be acknowledged. First, the temporal scope of the analysis was limited to the period between 2014 and 2024. While this range captures recent and relevant developments, it may exclude earlier foundational contributions and very recent emerging works. Second, the selection of articles relied primarily on the use of keywords. This approach, while systematic, did not distinguish journals according to their scientific quality or level of expertise, which may influence the overall representativeness of the sample. Third, the textual analysis conducted was largely based on the frequency and presence of terms rather than their full contextual meaning. As in any lexical study, this raises the possibility of misinterpretation, particularly in cases where terms may have been used ironically or in a nuanced context. Finally, the study remains secondary in nature, as it is based exclusively on previously published works

without access to the original data. This implies that any potential errors, omissions, or biases present in the primary studies are likely reproduced, with no possibility of correction within this research.

5.3. Recommendations for Future Research

Future research should focus on developing a specific implementation guide that explains how AI-based technologies can be concretely applied within tax administrations, as well as conducting empirical validation to assess the applicability and effectiveness of the proposed AATO framework in real institutional contexts. Another direction concerns the integration of AI within broader tax governance frameworks, particularly in the context of Tax Administration 3.0, to better understand how these technologies affect compliance costs, taxpayer trust, and the overall efficiency of tax systems.

In addition, comparative studies between high-income and low- and middle-income countries could shed light on the uneven adoption of AI tools in tax administration. Such work would highlight not only the technological and institutional challenges faced by emerging economies, but also identify context-specific strategies to ensure equitable and effective fraud detection. Finally, future research should explore the ethical and governance dimensions of AI adoption in tax systems, particularly issues of transparency, algorithmic bias, and taxpayer rights, in order to strengthen the legitimacy and sustainability of AI-driven tax oversight.

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